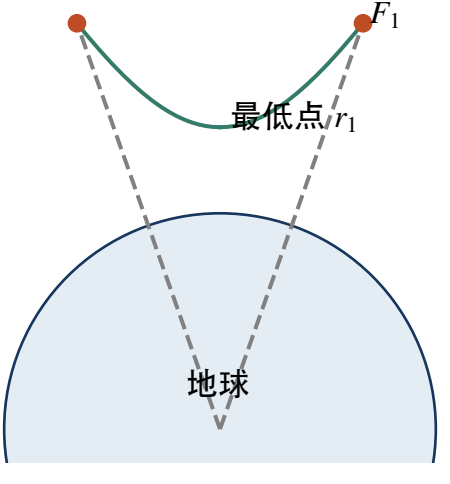
$$Tu\sin\alpha = F_2 r_1$$

$$T(u) = F_2 + \lambda g r^2 (1/r_1 - 1/u)$$